03. Feasify Functional Test Case Matrix

Document ID: FEASIFY-DOC-03

Scope: Functional Verification across Student, Adviser, and Chairperson Portals

Execution Date: November 11, 2026

Status: Completed & Verified

1. Test Case Format & Status Definitions

- **Passed:** Expected behavior matched actual observed behavior.
- **Failed:** Defect observed; deviates from expected requirements.
- **Blocked:** Precondition unmet or blocked by upstream issue.
- **Not Executed:** Safely bypassed due to environment boundaries.

2. Test Case Execution Matrix

2.1 Authentication & Account Management

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-AUTH-01	Auth	User Registration	1. Enter email, password, names. 2. Click "Create Account".	Creates user in Firebase Auth and writes profile to `users` collection.	User registered; profile created in `users` with `createdAt`.	**Passed**
TC-AUTH-02	Auth	Login Valid Credentials	1. Enter valid student email & password. 2. Click "Sign In".	Authenticates successfully and redirects to `/dashboard`.	Authenticated; redirected to `/dashboard`.	**Passed**
TC-AUTH-03	Auth	Login Invalid Credentials	1. Enter valid email with incorrect password. 2. Click "Sign In".	Displays error message: "Invalid email or password."	Error toast/alert displayed; authentication denied.	**Passed**
TC-AUTH-04	Auth	Session Persistence	1. Log in. 2. Refresh browser page.	Session maintained via Firebase Auth IndexedDB persistence.	User remains authenticated without re-login prompt.	**Passed**
TC-AUTH-05	Auth	Password Reset Request	1. Click "Forgot Password". 2. Enter registered email and submit.	Sends password reset email with action code URL.	`sendPasswordResetEmail` called with configured action URL.	**Passed**
TC-AUTH-06	Auth	User Sign Out	1. Open user dropdown. 2. Click "Sign Out".	Signs out user, clears local session, redirects to `/`.	`signOutUser` invoked; session terminated; redirected to `/`.	**Passed**

2.2 Role-Based Access Control (RBAC)

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-RBAC-01	RBAC	Student Access to Projects	1. Log in as Student. 2. Navigate to `/projects`.	Page rendered successfully; proposals loaded.	Rendered `Projects.tsx` view with student controls.	**Passed**
TC-RBAC-02	RBAC	Student Restricted from Admin	1. Log in as Student. 2. Force URL navigation to `/admin/users`.	`ProtectedRoute` blocks student and redirects to `/dashboard`.	Blocked by `ProtectedRoute`; redirected to `/dashboard`.	**Passed**
TC-RBAC-03	RBAC	Student Restricted from Adviser	1. Log in as Student. 2. Force URL navigation to `/adviser/dashboard`.	`ProtectedRoute` blocks student and redirects to `/dashboard`.	Blocked by `ProtectedRoute`; redirected to `/dashboard`.	**Passed**
TC-RBAC-04	RBAC	Adviser Access to Dashboard	1. Log in as Adviser. 2. Navigate to `/adviser/dashboard`.	Displays assigned student sections and proposal queue.	Displays assigned section tabs, filter controls, and proposals.	**Passed**
TC-RBAC-05	RBAC	Adviser Restricted from Chairperson	1. Log in as Adviser. 2. Force URL navigation to `/admin/users`.	`ProtectedRoute` blocks adviser and redirects to `/adviser/dashboard`.	Blocked; redirected to `/adviser/dashboard`.	**Passed**
TC-RBAC-06	RBAC	Chairperson Elevated Access	1. Log in as Chairperson. 2. Navigate across `/admin/` *and* `/adviser/`.	Full access granted to both admin and adviser modules.	Permitted by `ProtectedRoute` (`allowedRoles: ["Chairperson"]`).	**Passed**
TC-RBAC-07	RBAC	Hardcoded Chairperson Email Bypass	1. Inspect `ProtectedRoute.tsx` L31 and `index.js` L97. 2. Test login with `chairperson@gmail.com`.	Role is assigned strictly from Firestore profile record.	**VULNERABILITY FOUND:** Hardcoded email grants instant Chairperson role without DB check.	**Failed**

2.3 Capstone Proposal Management & 3-Proposal Limit (10 Scenarios)

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-PROP-01	Proposal	Limit Scenario 1: Zero Proposals	1. New group created. 2. Query `/api/teams/:id/proposals/count`.	Count = 0, Max = 3, Allowed = true.	Count = 0, Max = 3, Allowed = true (`HTTP 200`).	**Passed**
TC-PROP-02	Proposal	Limit Scenario 2: One Proposal	1. Submit Proposal 1. 2. Query proposal count.	Count = 1, Max = 3, Allowed = true.	Proposal 1 created (`proposalNumber: 1`); Count = 1.	**Passed**
TC-PROP-03	Proposal	Limit Scenario 3: Two Proposals	1. Submit Proposal 2. 2. Query proposal count.	Count = 2, Max = 3, Allowed = true.	Proposal 2 created (`proposalNumber: 2`); Count = 2.	**Passed**
TC-PROP-04	Proposal	Limit Scenario 4: Three Proposals	1. Submit Proposal 3. 2. Query proposal count.	Count = 3, Max = 3, Allowed = false.	Proposal 3 created (`proposalNumber: 3`); Count = 3, `allowed: false`.	**Passed**

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-PROP-05	Proposal	Limit Scenario 5: Attempt Fourth Proposal	1. With 3 proposals on file, attempt to submit Proposal 4.	Client blocks form; backend rejects with `HTTP 400` ("Maximum of 3 proposals reached").	Blocked on client modal; backend returns `HTTP 400` rejection.	**Passed**
TC-PROP-06	Proposal	Limit Scenario 6: Browser Refresh Persistence	1. At 3 proposals, refresh page. 2. Attempt to add new proposal.	State persists from database; Add button disabled or modal displayed.	Count persists as 3; allowed remains false; creation blocked.	**Passed**
TC-PROP-07	Proposal	Limit Scenario 7: Direct API Manipulation Bypass	1. Send direct `POST /api/proposals` payload with group ID having 3 proposals.	Backend rejects with `HTTP 400` and `proposalLimitReached: true`.	Rejected with `HTTP 400` and `proposalLimitReached: true`.	**Passed**
TC-PROP-08	Proposal	Limit Scenario 8: Simultaneous Concurrent Submissions	1. Fire 5 concurrent creation requests for the same group ID.	Exactly 3 succeed (`HTTP 201`); remaining 2 rejected (`HTTP 400`).	Exactly 3 created; 2 rejected with `HTTP 400`. Race condition safely handled.	**Passed**
TC-PROP-09	Proposal	Limit Scenario 9: Edit Existing Proposal at Limit	1. With 3 proposals on file, edit and save changes to Proposal 2.	Editing existing proposal succeeds; does not increment total count.	Existing proposal updated (`HTTP 200`); total count remains 3.	**Passed**
TC-PROP-10	Proposal	Limit Scenario 10: Status Counting Rules	1. Check whether Drafts, Under Review, and Rejected proposals count toward limit.	All proposals belonging to team count toward the 3-proposal cap.	All proposals for `groupId` count toward the 3-proposal limit.	**Passed**
TC-PROP-11	Proposal	Direct Client Firestore Bypass Risk	1. Inspect `Projects.tsx` L1793. 2. Check if client can bypass backend via direct Firestore write.	All creation requests must pass through backend validation.	**VULNERABILITY FOUND:** Client writes directly to Firestore via `addDoc`, bypassing backend if Firestore rules are permissive.	**Failed**

2.4 Proposal Submission Button & Form Validation

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-FORM-01	Submission	Incomplete Form Submission	1. Leave required fields blank (e.g., Target Market, Location). 2. Click "Submit to Adviser".	Submission blocked; modal displays exact missing field labels.	Blocked; `SubmissionFailureModal` lists missing fields with bullet points.	**Passed**
TC-FORM-02	Submission	Negative Capital Validation	1. Enter `-50000` in Total Capital. 2. Attempt to save or submit.	Blocked with toast: "Total capital cannot be negative."	Blocked; capital validation catches negative number and alerts user.	**Passed**

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-FORM-03	Submission	Non-Leader Submission Check	1. Log in as a team member (not leader). 2. Attempt to add or submit proposal.	Blocked with toast: "Only the team leader is capable of adding a new proposal."	Action blocked; alert displayed informing user of leader requirement.	**Passed**
TC-FORM-04	Submission	Loading State & Double-Click Protection	1. Fill form and click "Submit to Adviser".	Button shows spinner (`isSubmitting: true`); disables double-clicks.	Spinner rendered; button disabled during async transaction.	**Passed**

2.5 Adviser & Chairperson Evaluation Workflows

Test Case ID	Module	Test Scenario	Steps	Expected Result	Actual Result	Status
TC-ADV-01	Adviser	Proposal Review View	1. Log in as Adviser. 2. Click on a submitted student proposal.	Full proposal details, financials, and remarks input load.	Details modal renders all sections and financial summary.	**Passed**
TC-ADV-02	Adviser	Approve Proposal	1. Enter remarks. 2. Click "Approve Proposal".	Status updates to "Approved"; student group status updates; notif sent.	Status changed to "Approved"; proposal marked active for group.	**Passed**
TC-ADV-03	Adviser	Request Revision	1. Enter remarks explaining required changes. 2. Click "Request Revision".	Status updates to "Revision Required"; remarks logged in history.	Status changed to "Revision Required"; student notified to revise.	**Passed**
TC-ADV-04	Adviser	Reject Proposal	1. Enter justification. 2. Click "Reject Proposal".	Status updates to "Rejected"; proposal disabled from active setup.	Status changed to "Rejected"; feedback history recorded.	**Passed**
TC-ADV-05	API	Unauthenticated Status Endpoint Vulnerability	1. Issue `PUT /api/proposals/:id/status` without Authorization header.	Endpoint rejects request with `HTTP 401 Unauthorized` .	**VULNERABILITY FOUND:** Request succeeds with `HTTP 200` ; missing `verifyToken` middleware.	**Failed**